

How the Van de Graaff Generator Works

Step 1. At the Bottom: Creating the Charge

- **Friction:** The motor spins, rubbing the **rubber belt** against a **silicon roller**.
- **Electron Transfer:** The belt gives its electrons to the roller because rubber has a lower electron affinity than silicon
- **The Result:** The belt becomes **positively charged** as it heads toward the top.

Step 2. At the Top: Collecting the Charge

- **The Goal:** The positive belt "wants" to be neutral again, so it needs to find electrons.
- **The Swap:** As the belt passes the top metal brush/dome combination and nylon roller, it pulls electrons from the metal brush/dome combination and the nylon roller.
- **The Result:** Because the brush/dome is losing its negative electrons, it becomes **positively charged**.

Step 3. Why does your hair stand up?

- **The Handshake:** You start out as neutral because you haven't been charged by anything. When you touch the dome, electrons from your body jump onto the dome to try and neutralize it.
- **Becoming Positive:** Losing electrons to the dome makes you (and your hair) positively charged. Since the generator is run by a motor, it can't get to neutral because it is constantly being recharged. So, the dome remains positive
- **The Big Repel:** Since "like" charges repel, your positive hair strands try to get as far away from the positive dome—and each other—as possible. This makes your hair lift and spread out!